

Objective

To implement and demonstrate Interface Inheritance in Java, where interface I1 extends both I2 and I3, and class C1 implements I1.

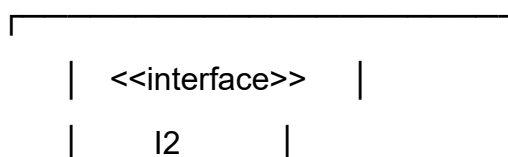
Steps for Execution

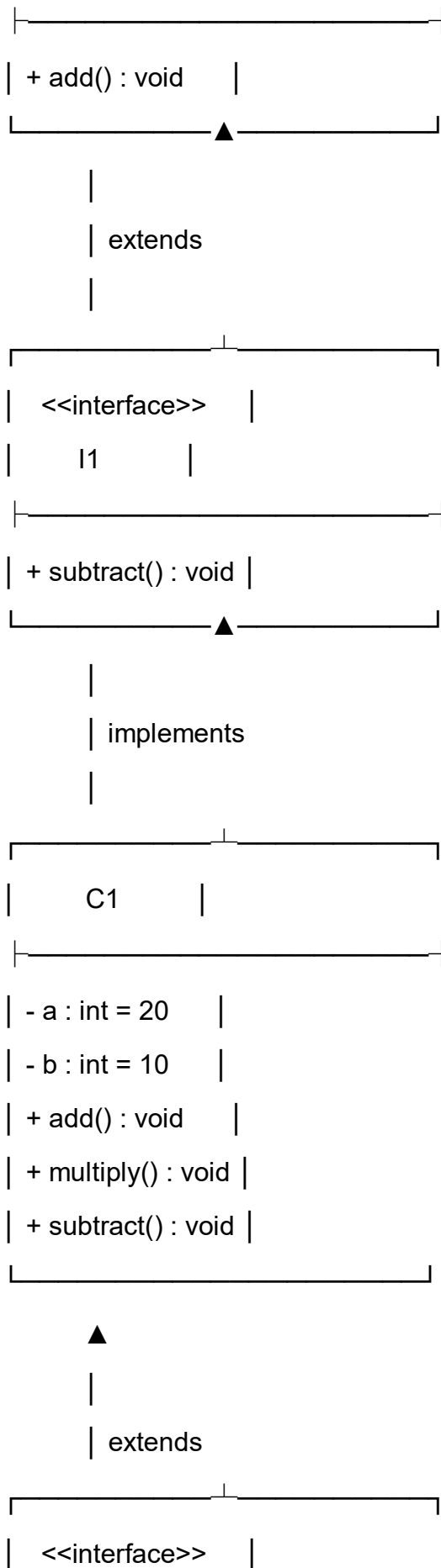
1. Create a Java project and package named javacore.
2. Create a file named Main.java and enter the given code.
3. Define interfaces I2 and I3 with add() and multiply() methods.
4. Define interface I1 extending I2 and I3.
5. Create class C1 implementing I1 and define all three methods.
6. Create a C1 object and call the three methods.
7. Compile and run the program.
8. Observe the output.

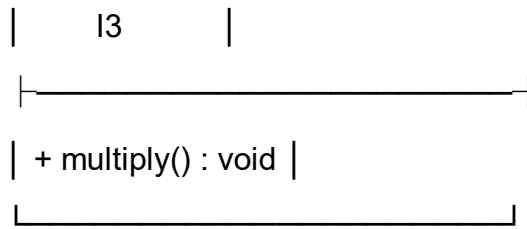
Algorithm

1. Start the program.
2. Define interfaces I2 and I3.
3. Define I1 extending I2 and I3.
4. Create class C1 implementing I1.
5. Implement add(), multiply(), and subtract().
6. Create a C1 object.
7. Call all three methods.
8. Display the output.
9. Stop the program.

UML CLASS DIAGRAM:







Relationships:

- I1 extends I2
- I1 extends I3
- C1 implements I1
- C1 implements add(), multiply(), and subtract().

CODE:

```
package javacore;
```

```
interface I2 {  
    void add();  
}
```

```
interface I3 {  
    void multiply();  
}
```

```
interface I1 extends I2, I3 {  
    void subtract();  
}
```

```
class C1 implements I1 {
```

```
    int a = 20;
```

```
    int b = 10;
```

```
    @Override
```

```
public void add() {  
    System.out.println("Addition = " + (a + b));  
}  
  
@Override  
public void multiply() {  
    System.out.println("Multiplication = " + (a * b));  
}  
  
@Override  
public void subtract() {  
    System.out.println("Subtraction = " + (a - b));  
}  
}
```

```
public class Main {  
    public static void main(String[] args) {  
  
        C1 obj = new C1();  
  
        obj.add();  
        obj.multiply();  
        obj.subtract();  
    }  
}
```

OUTPUT:

Addition = 30

Multiplication = 200

Subtraction = 10

Result:

The program successfully demonstrates Interface Inheritance in Java. Interface I1 inherits methods from both I2 and I3, while class C1 implements I1 and provides implementations for all three methods.